

DATA

ANALYTICS PORTFOLIO

Name of Project:
Data Analytics Portfolio

Presented By:
Hena Delic



Hello, I'm Hena Delic!

I am a second year Masters in Data Science student at Boston University. I spent the summer of 2026 as a Data Analytics and Marketing intern at Starting Point NEO, a Child Care Resource and Referral Agency in Cleveland, OH. I was placed here as part of the Cleveland Foundation's summer internship program. I learned a lot during my time at Starting Point and wanted to share the projects I worked on.

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Summer Project



My main internship project this summer involved extracting, transforming and loading data from datasets across several programmatic areas at Starting Point. I then analyzed this data for major patterns in Microsoft Excel and used data visualization tools like PowerBI to convey my findings.

Goals & targets

My goal was to show that the work being done at Starting Point to support families and child care centers to offer the best care to young children is not only working but also driving change in the community.

Analytics & Reports

My analytics will be used in this year's annual report. Additionally, I collaborated with the marketing team to film a [video series](#) showcasing my work which is featured across their social media.

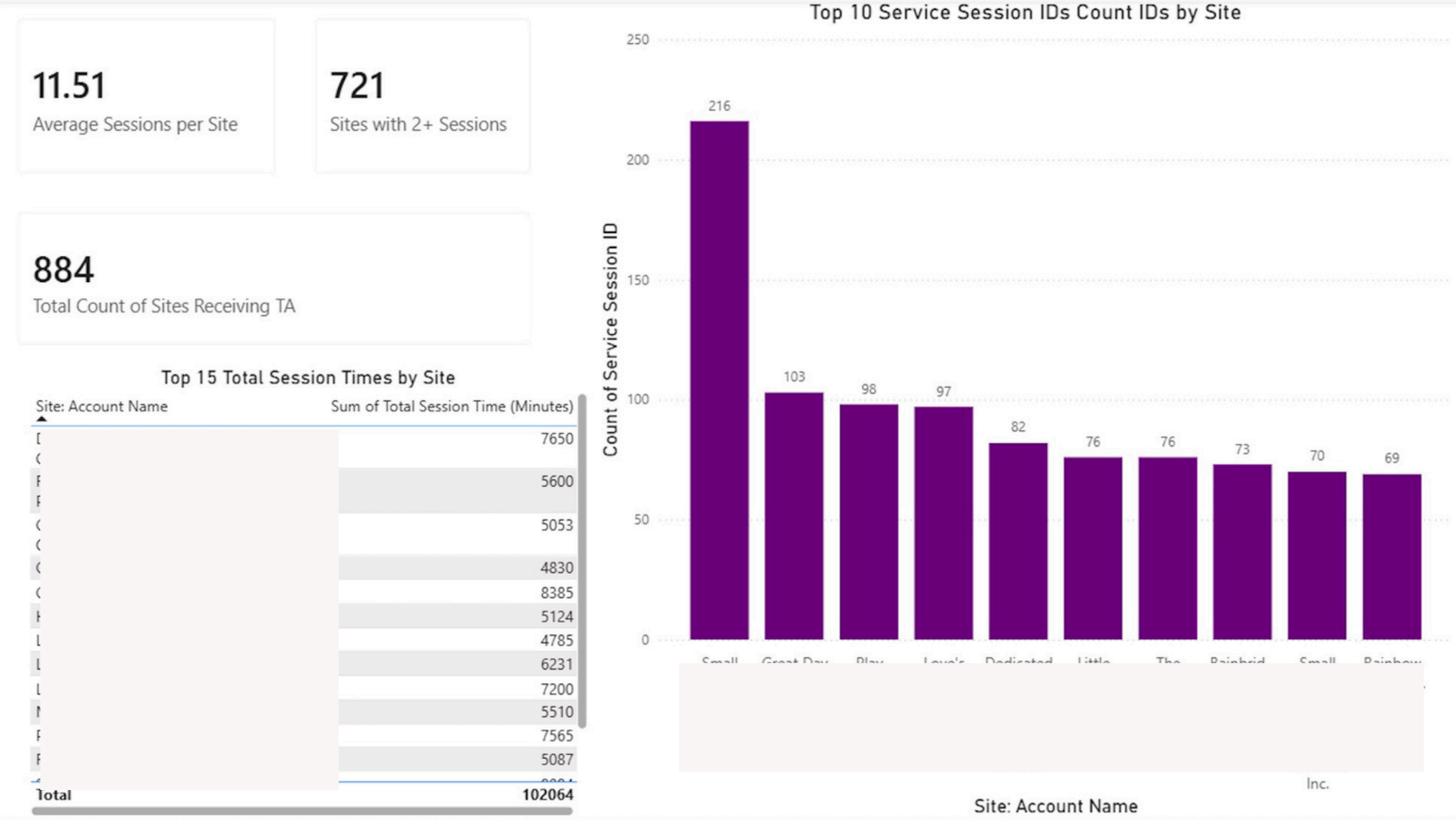
Adaptation & Evolution

As the first person to analyze some of these datasets, my work will serve as a foundation for future analytics at the organization.



Part I: Technical Assistance

My goal was to compare how different child care centers performed before and after receiving Technical Assistance. Step Up to Quality is a three-star rating system used by the state of Ohio to measure child care center quality. I wanted to know if TA was helping childcare centers in Northeast Ohio improve their SUTQ ratings. After doing preliminary data exploration I used Excel to identify child care center's first and last SUTQ ratings and which types of TA were driving the most change.



The service types with the most TA sessions in 2024-2025 were Administration, DECA, Learning and Development and Quality Enhancement, all of which had over 1,000 sessions. Total sessions for all service types was 10,326 meaning those four made up about 70% of TA sessions across 886 providers.

10326

Total TA Sessions Across Services

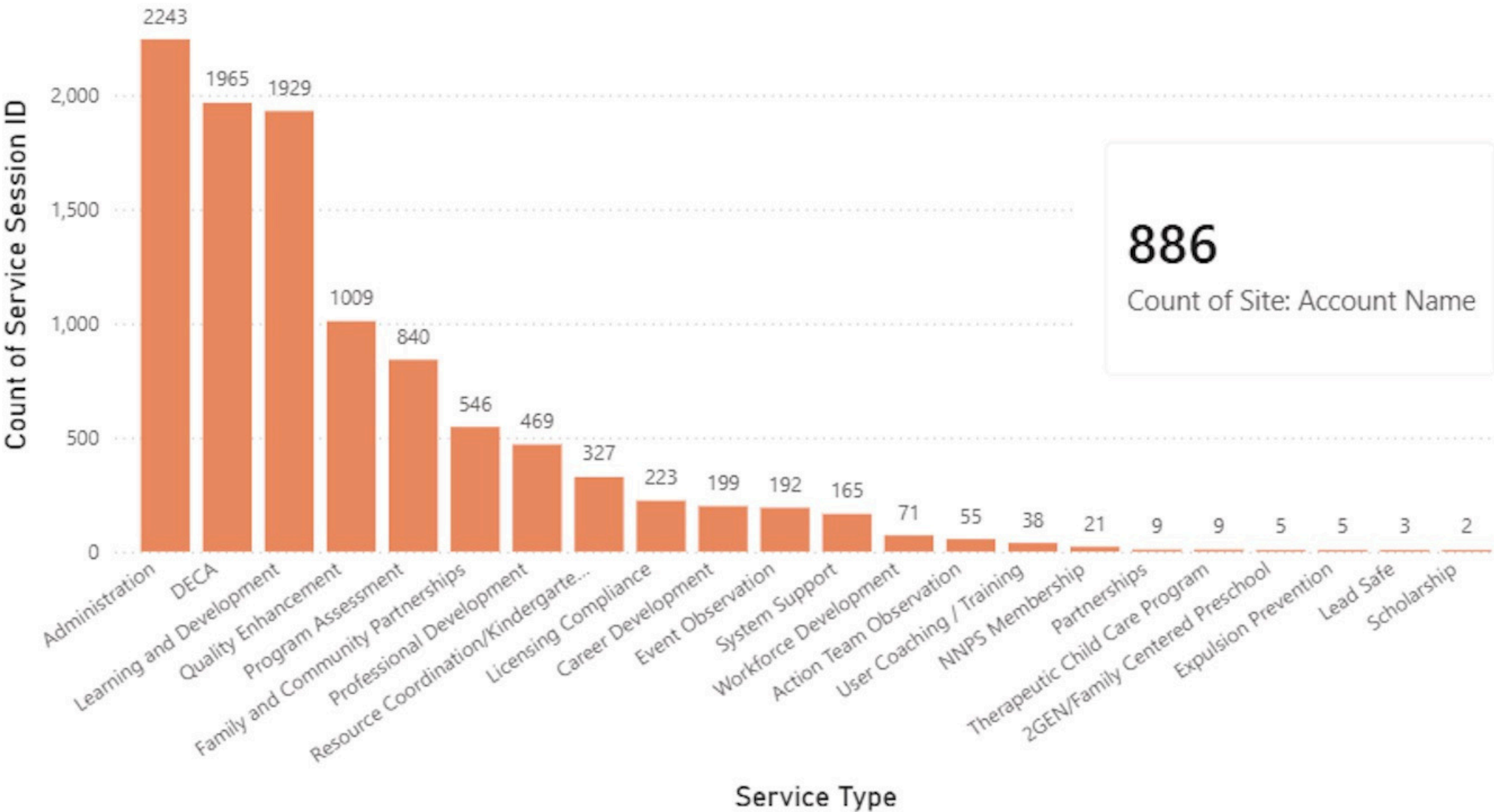
23

Total Count of Service Types

134

Median Sessions by Service Type

Count of TA Service Session ID by Service Type



886

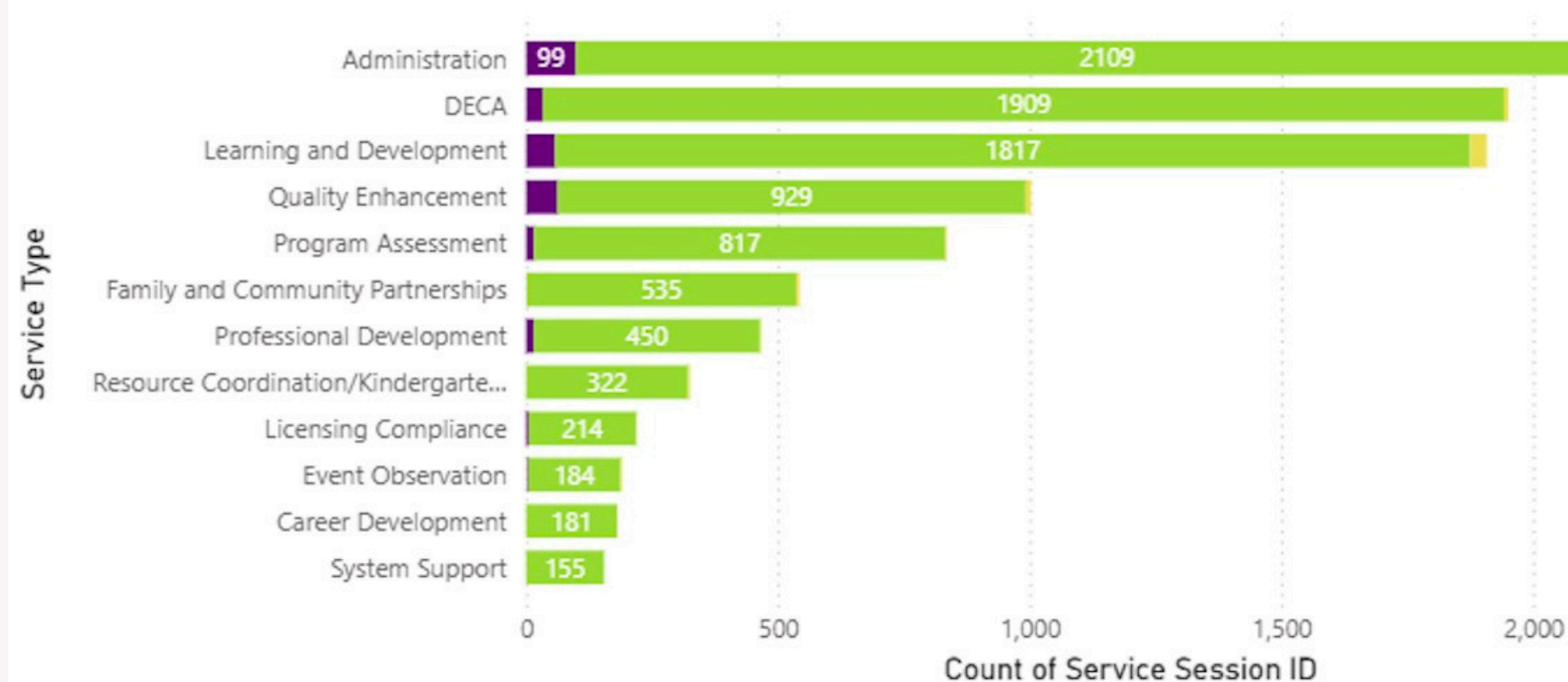
Count of Site: Account Name

TA Session Count by Site

Site: Account Name	Count of Service Session ID
	216
	147
	103
	98
	97
	82
	76
	76
	73
	70
	69
	68
	65
Total	10326

Count of Service Session ID by Service Type and Outcome

Outcome ● Cancelled ● Completed ● No Show



Count of Completed TA Sessions by Service Type

Service Type	Count of Completed Session
Administration	2109
DECA	1909
Learning and Development	1817
Quality Enhancement	929
Program Assessment	817
Family and Community Partnerships	535
Professional Development	450
Resource Coordination/Kindergarten Transition	322
Licensing Compliance	214
Event Observation	184
Career Development	181
System Support	155
Workforce Development	6
Action Team Observation	5
User Coaching / Training	3
NNPS Membership	2
Partnerships	0
Therapeutic Child Care Program	0
2GEN/Family Centered Preschool	0
Expulsion Prevention	0
Scholarship	0
Lead Safe	0
Total	9830

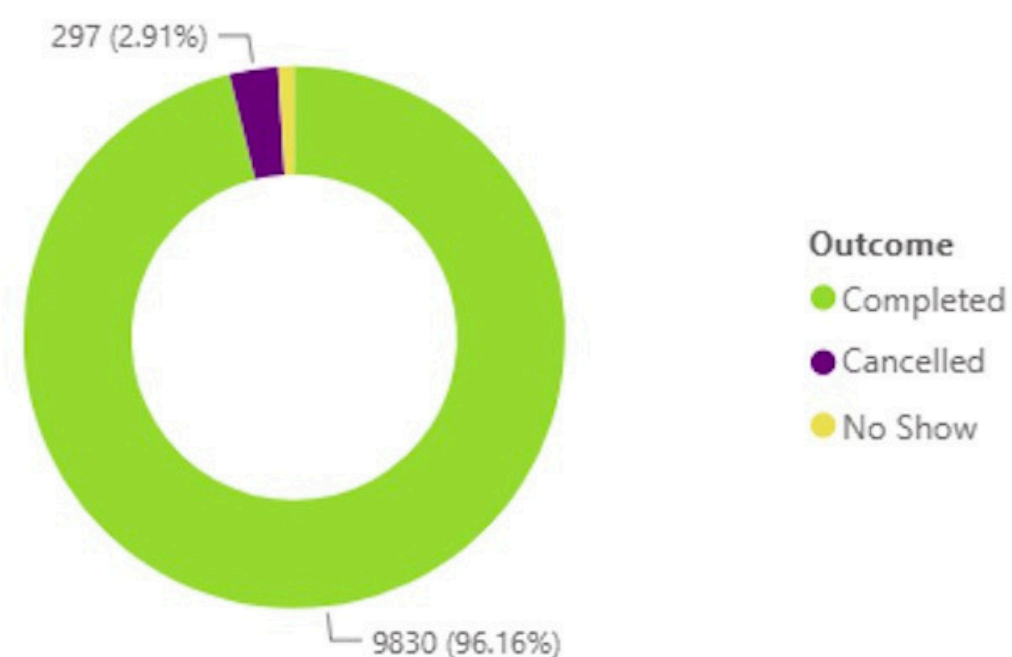
95.20%

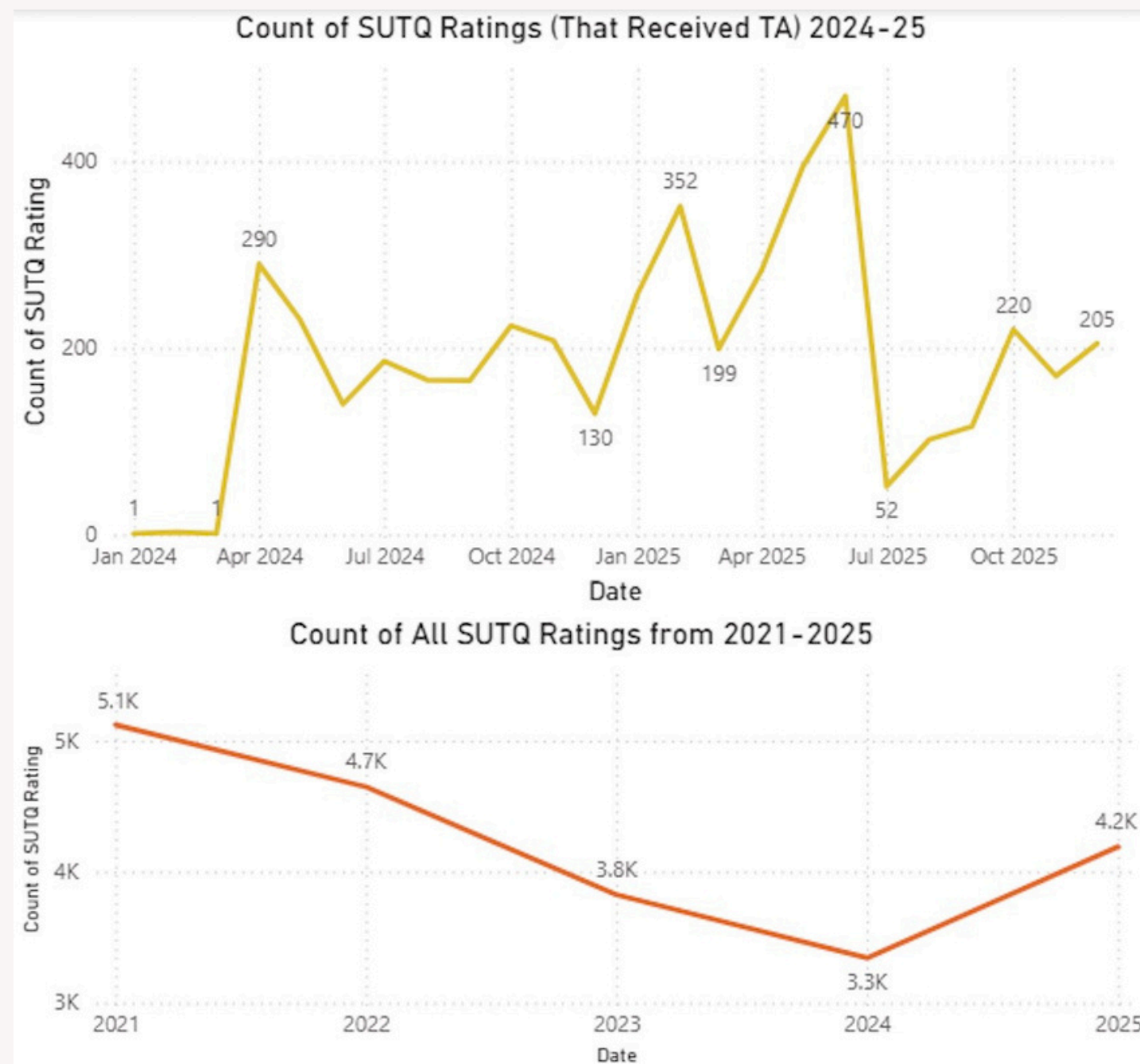
TA Session Outcome Completion Rate

2.88%

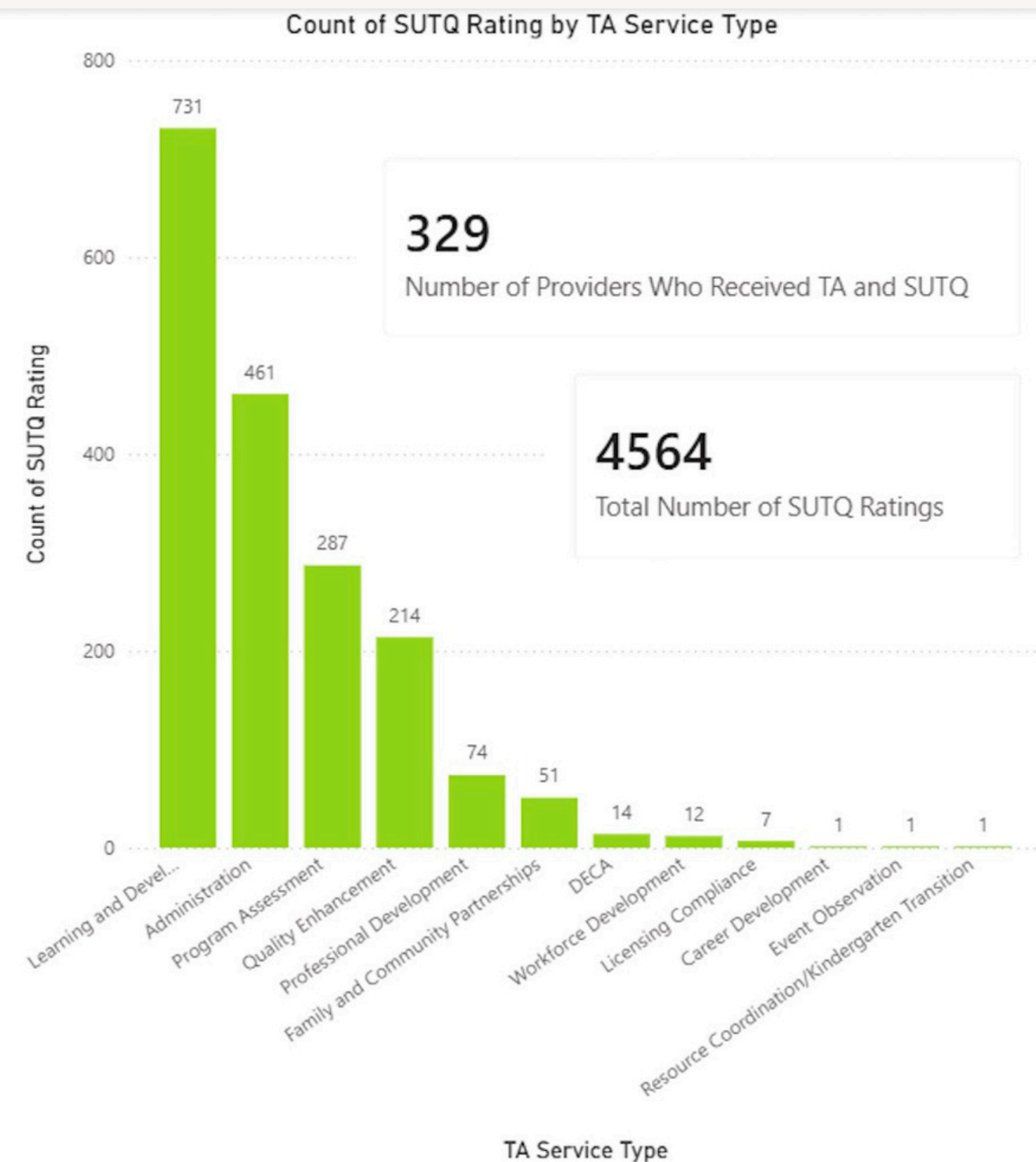
TA Session Cancellation Rate

Count of TA Sessions by Outcome





In 2020, the pandemic and distrust towards funding consistency [disrupted participation rates](#). However, in 2024 [SUTQ changed](#) from a five star system to a bronze/silver/gold tier system. This reduced administrative burden and increased parental access, resulting in an increase in participation across providers in NEO.



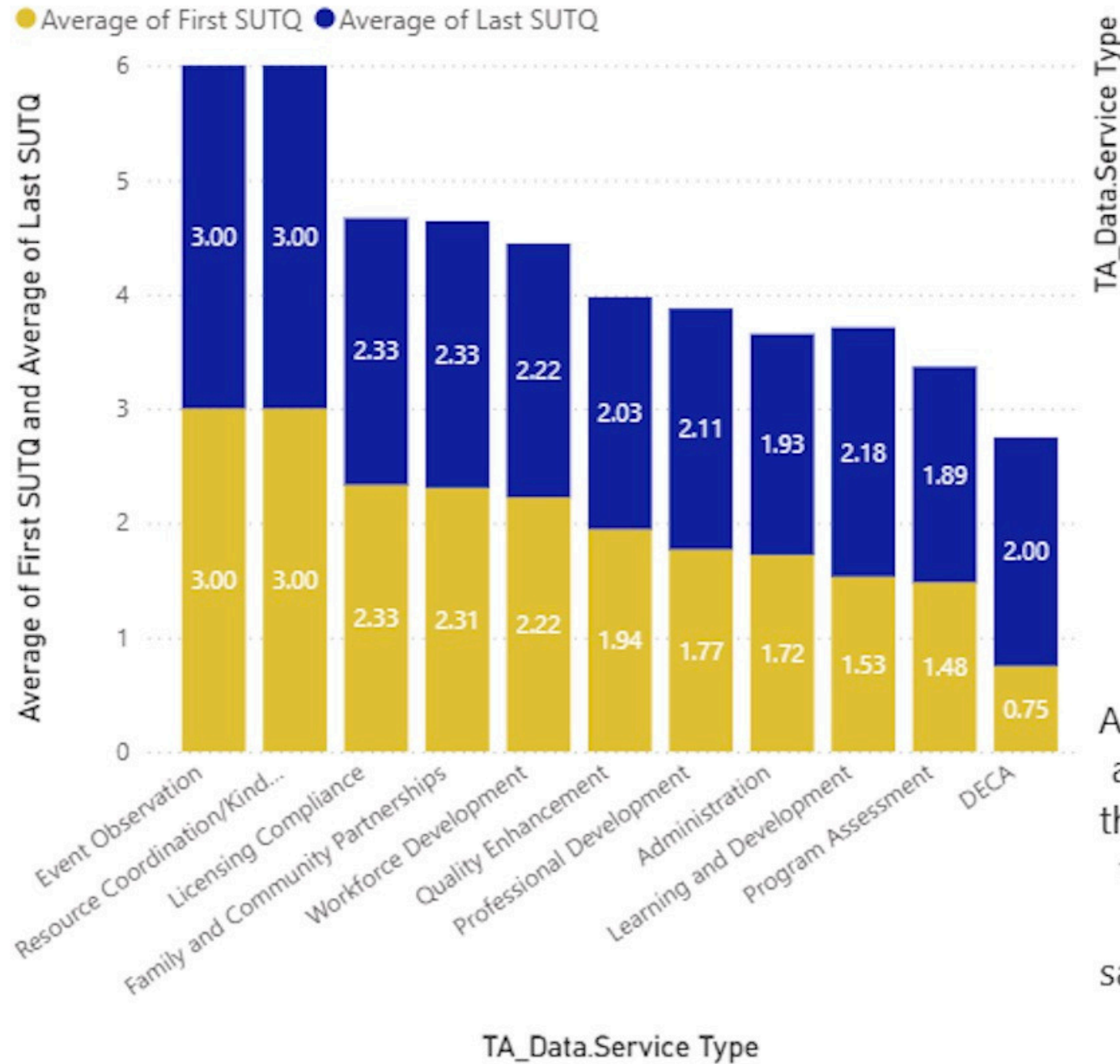
1.69

Average of First SUTQ

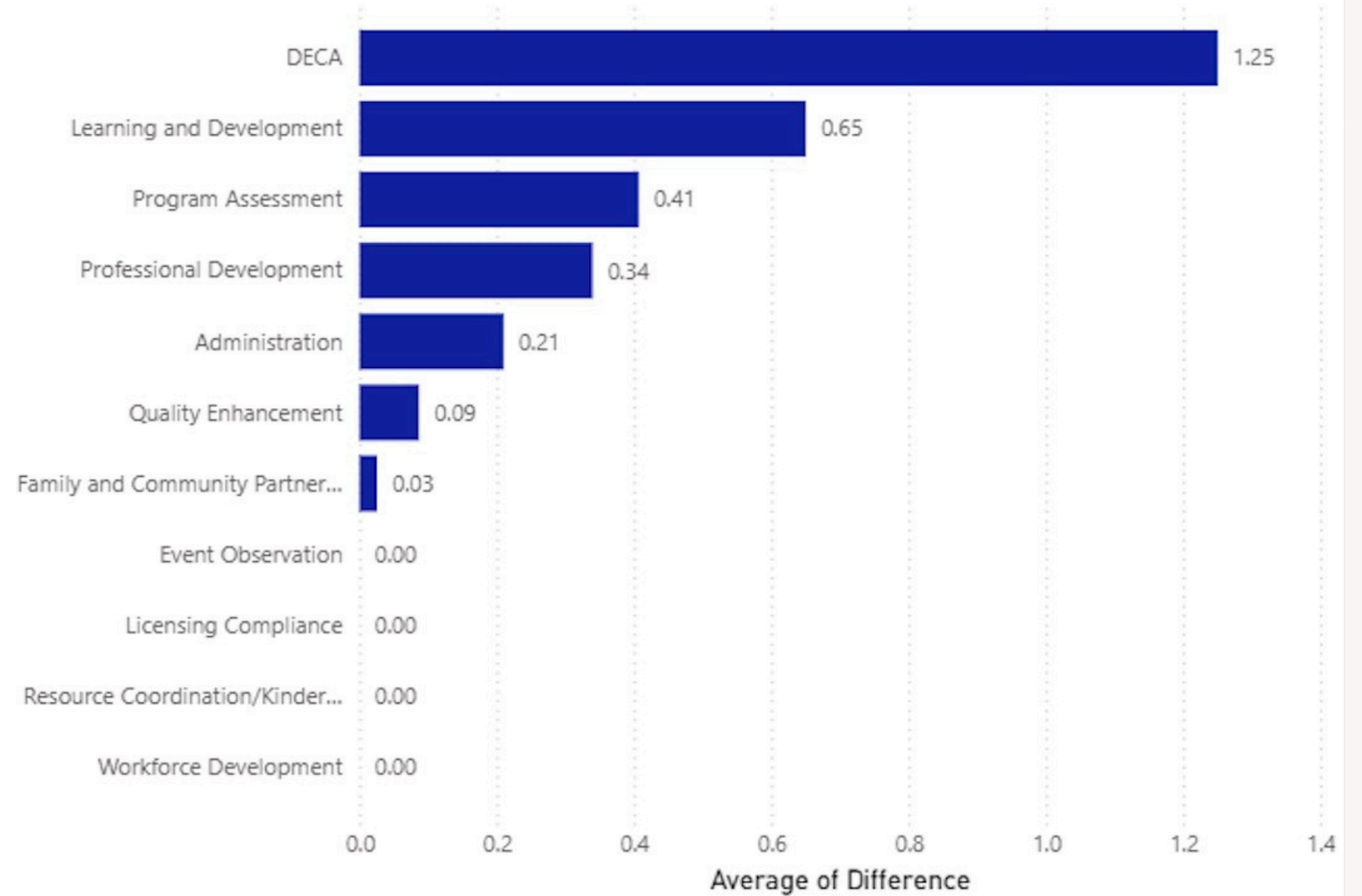
2.03

Average of Last SUTQ

Average of First & Last SUTQ Rating by TA Service Type



Which TA Services Resulted in SUTQ Rating Improvement?



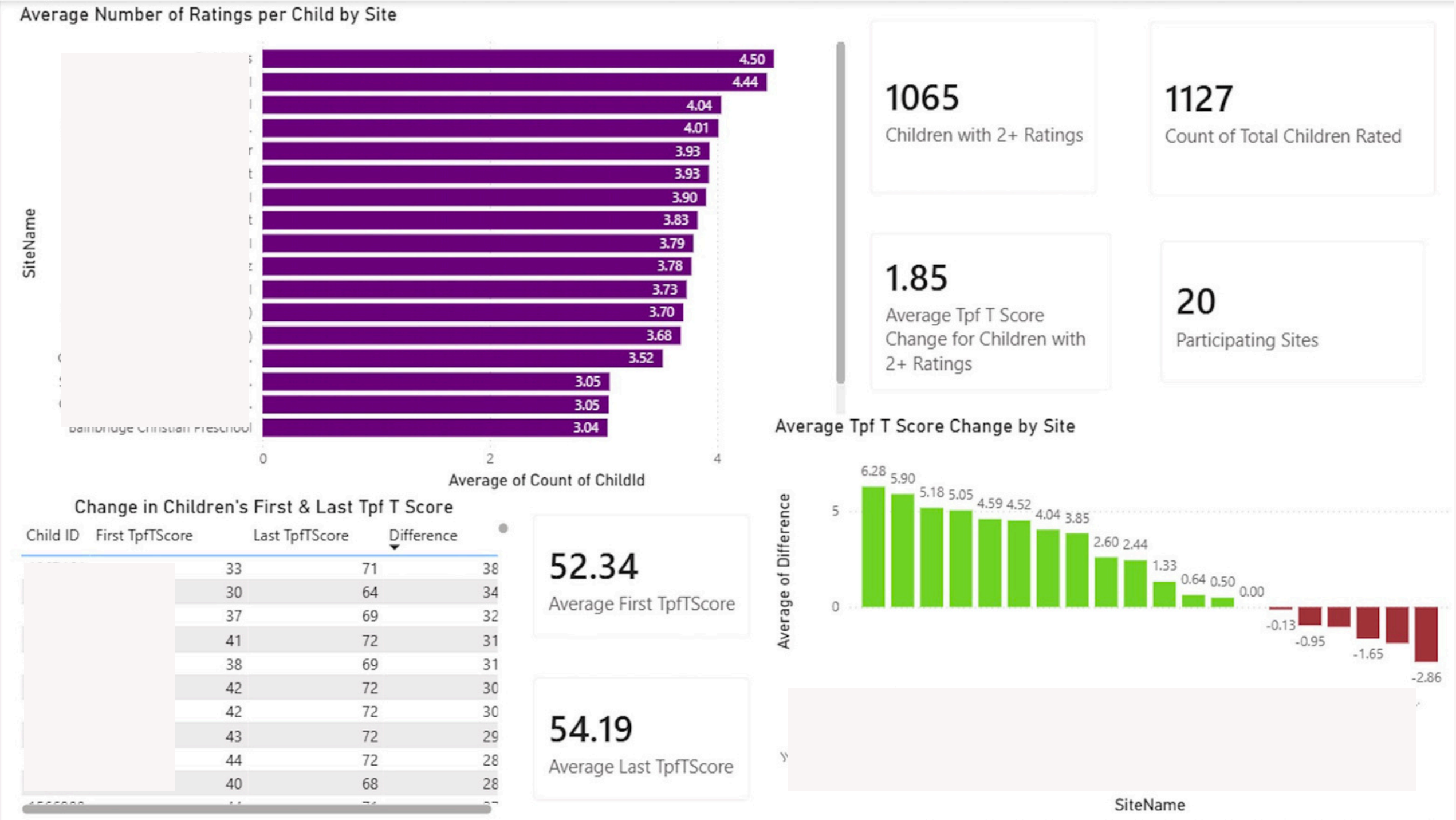
All TA Service Types have an average improvement rate that is greater than or equal to zero. 97.39% of ratings improved or stayed the same for those seeking new or higher SUTQ.

How Many TA Services Resulted in SUTQ Improvement?

Change?	Count of Count of Encounter ID	%GT Count of Count of Encounter ID
Declined	64	3.61%
Improved	390	21.98%
No Change	1320	74.41%
Total	1774	100.00%

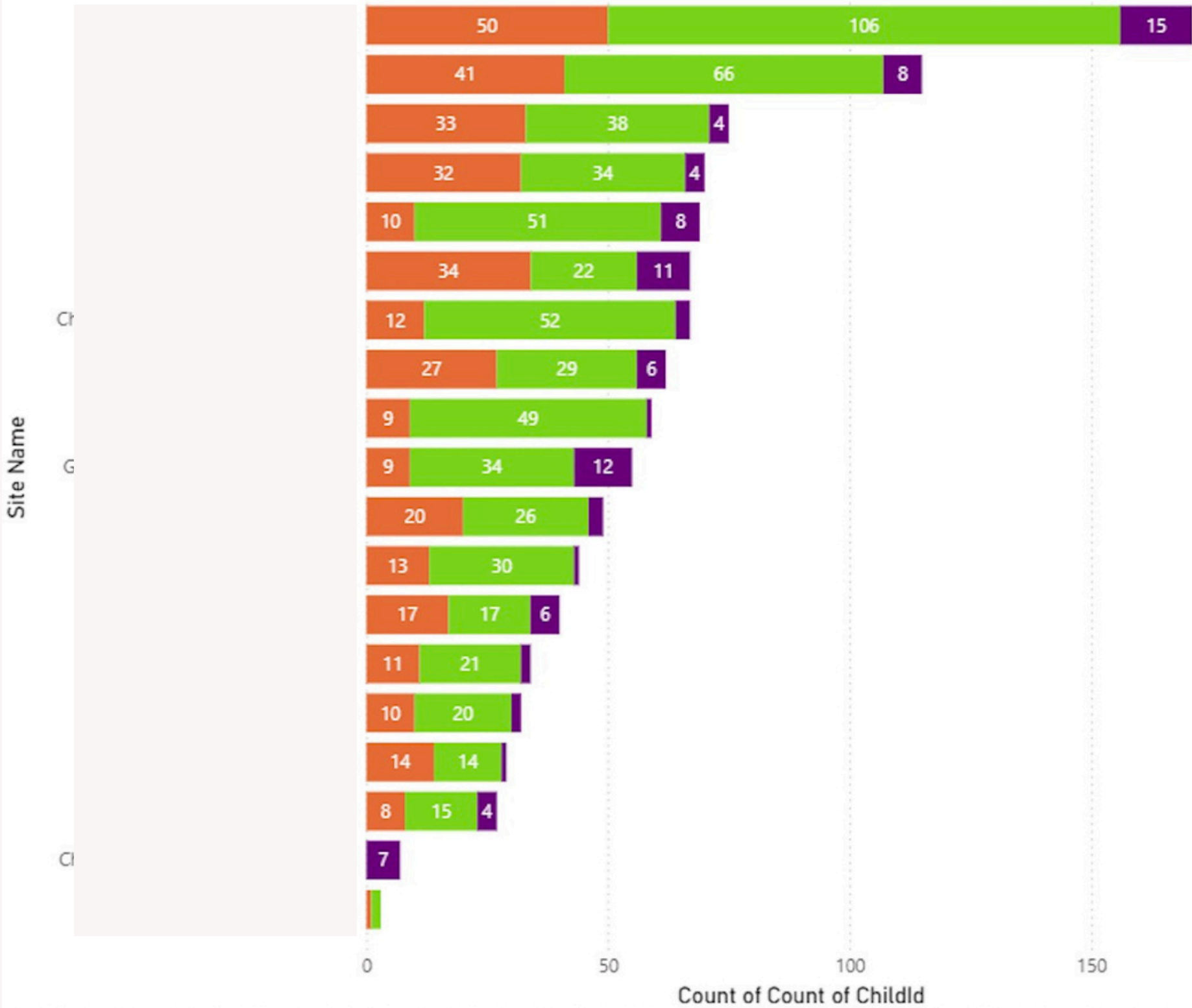
Part II: Devereux Early Childhood Assessment

My goal was to compare how children across different child care centers performed developmentally before and after undergoing evaluation by an early childhood development assessment called DECA. TPF stands for total protective factors and is the most comprehensive measure which is why I analyzed t-scores from this category. I then, using Excel, calculated each child's (with 2+ sessions) score improvement and categorized them by Improved, Declined and No Change and compared which sites had the highest improvement rates to determine where children are performing the strongest.



Count of Count of Childld by Site Name and Improvement?

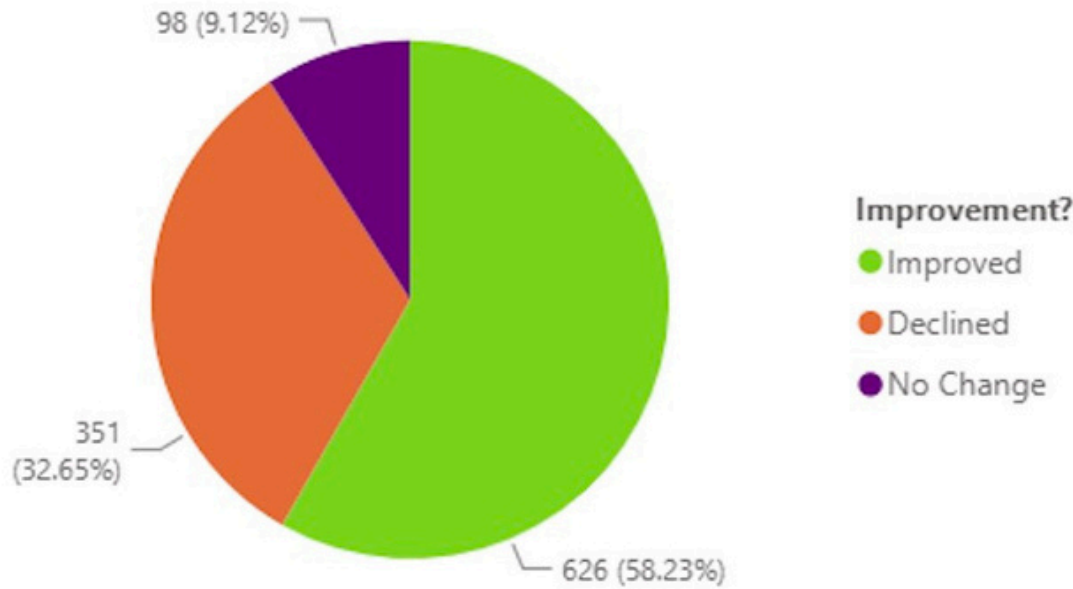
Improvement? Declined Improved No Change



Improvement Percentage by Site Name

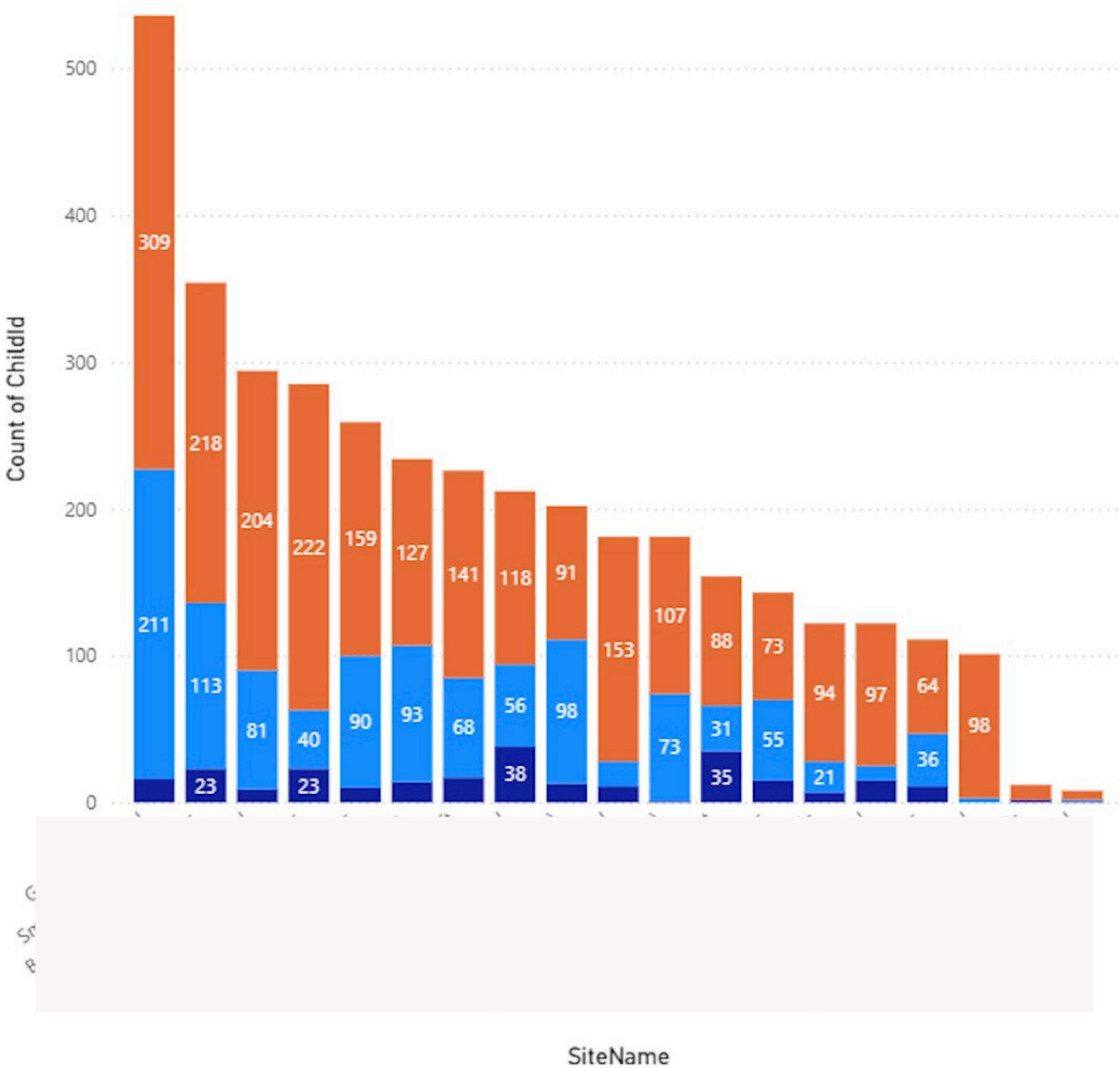
Site Name	Declined	Improved	No Change
	44.44%	35.56%	20.00%
	38.89%	52.78%	8.33%
r	17.02%	76.60%	6.38%
	9.09%	81.82%	9.09%
	10.00%	85.00%	5.00%
	27.93%	65.77%	6.31%
er	15.63%	59.38%	25.00%
	20.00%	70.00%	10.00%
	48.00%	48.00%	4.00%
	9.38%	81.25%	9.38%
	19.23%	80.77%	
	29.41%	70.59%	
	20.00%	53.33%	26.67%
	54.29%	42.86%	2.86%
	44.74%	53.95%	1.32%
Total	30.71%	61.43%	7.86%

Count of Children's Tpf T Score Improvement Across All Sites



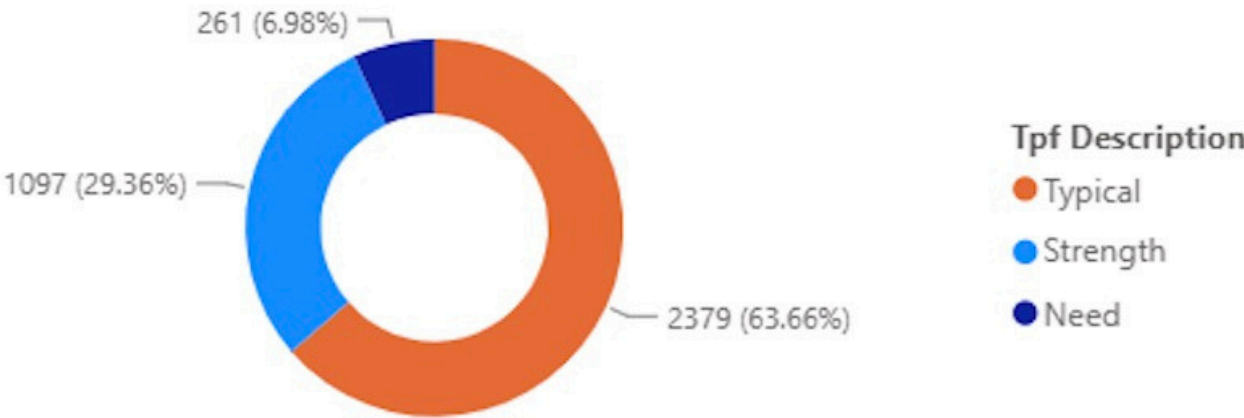
Count of ChildId by SiteName and Tpf Description

Tpf Description ● Need ● Strength ● Typical



SiteName	Average of TpfTScore	Average of TpfPercentile
	56.08	66.45
	54.79	65.24
	53.13	61.00
	55.24	65.70
	51.91	57.02
	50.82	52.42
	57.05	70.13
	54.82	62.94
	51.76	54.02
	51.43	54.46
	59.65	74.42
	58.28	73.30
	52.92	59.34
	49.31	47.39
	54.50	62.48
Total	54.50	62.74

Count of ChildId by TpfDescription Across all Sites



62.74

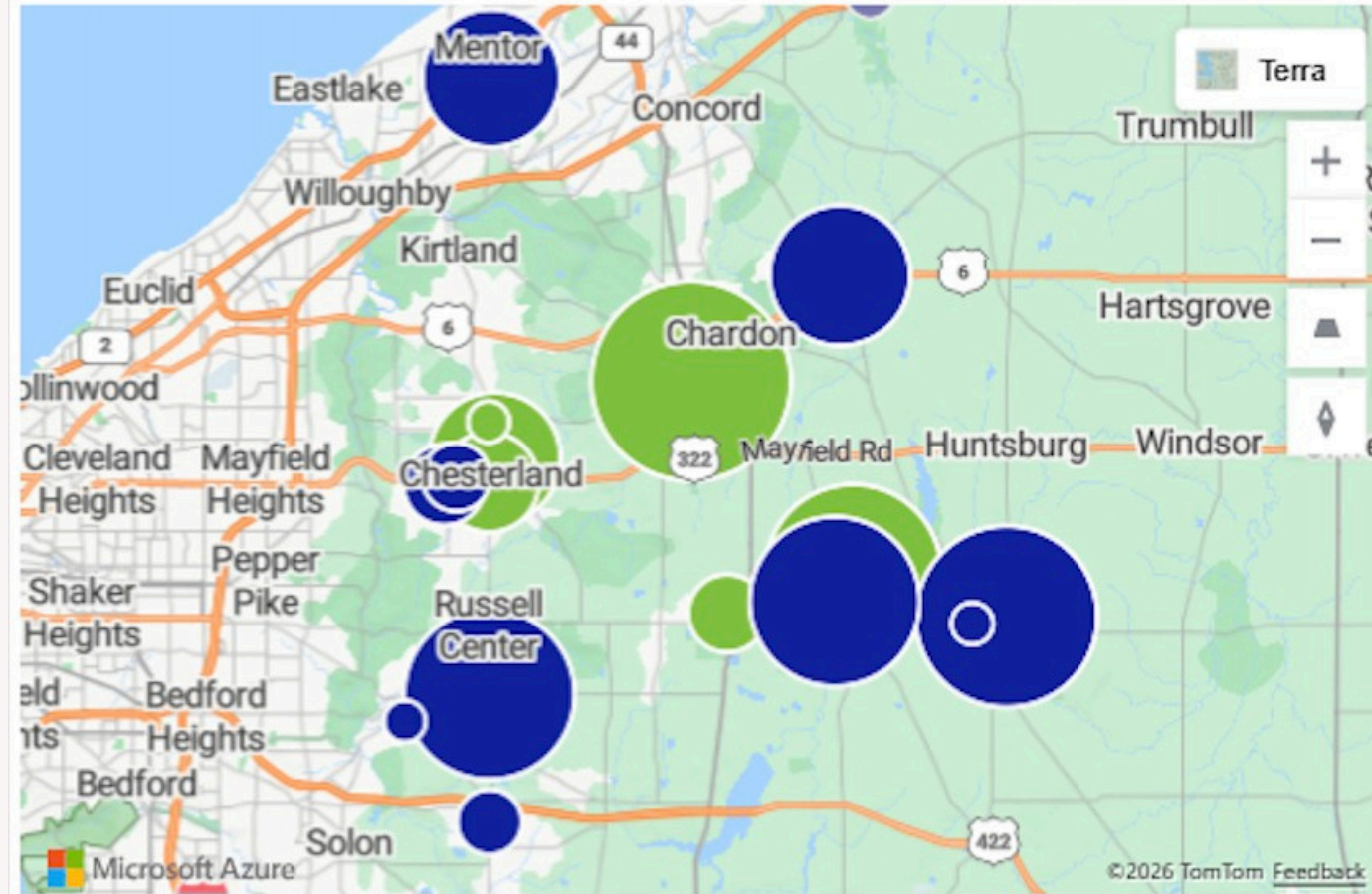
Average TpfPercentile for all Sites

54.50

Average TpfTScore for all Sites

Average of TPF Score Difference by Site (Geauga 24-25)

Provider Type ● Preschool Program ONLY ● Child Care Center



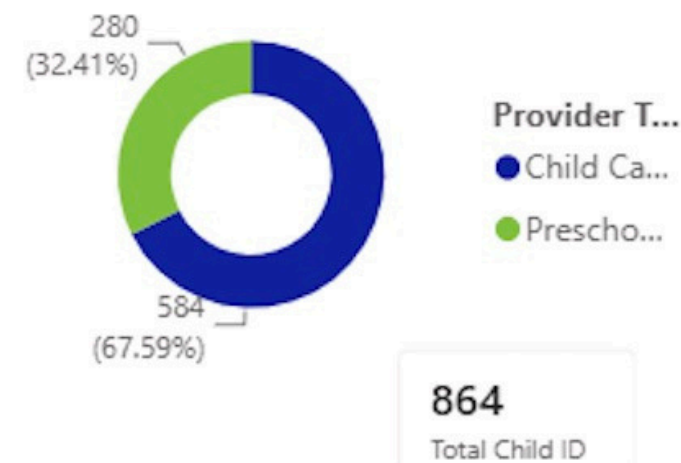
7

Count of Preschool Programs

10

Count of Child Care Centers

Count of Child ID by Provider Type

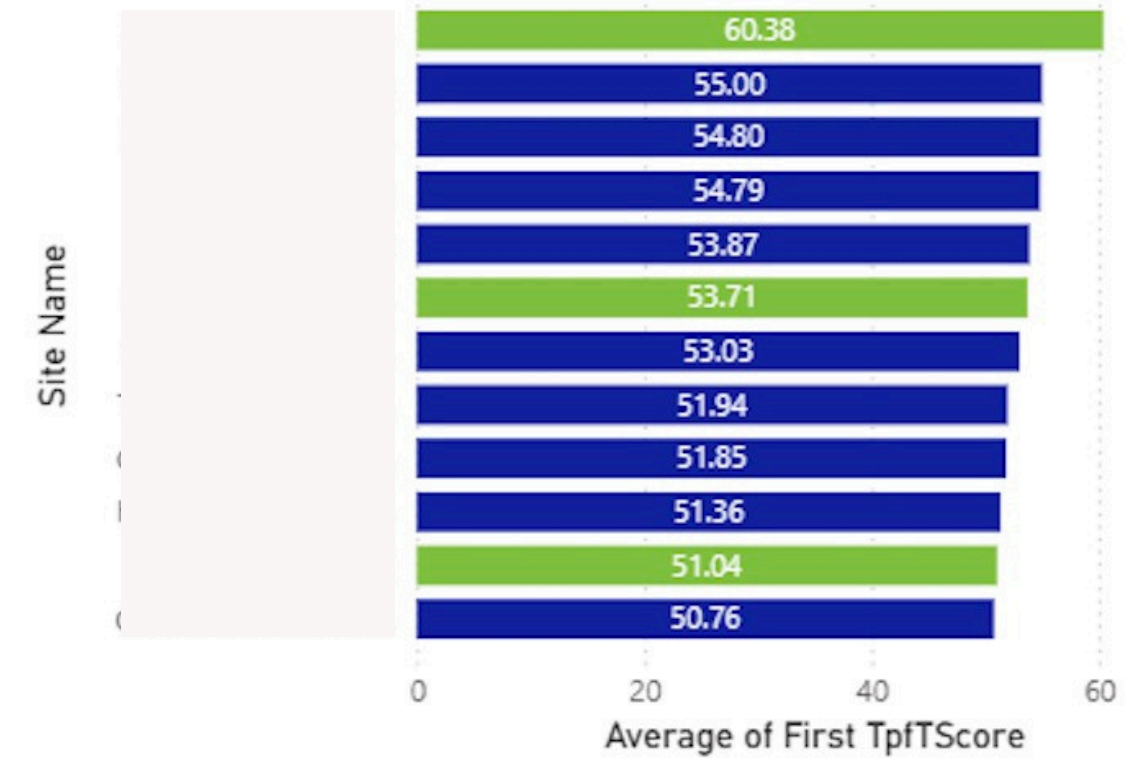


Change in TPF T Score by City

City	Average of First TpfTScore	Average of Last TpfTScore	Average of Difference	Count of Providers	Count of ChildId
Chardon	52.44	60.65	8.21	2	91
Chardon	57.59	64.95	7.36	2	103
Chardon	51.85	58.81	6.96	1	67
Chardon	53.03	56.90	3.88	1	115
Chardon	52.98	56.07	3.10	2	122
Chardon	51.68	52.84	1.16	5	206
Chardon	53.40	54.07	0.66	3	131
Total	53.03	56.66	3.63	17	864

Average of First TpfTScore by Site Name and Provider Type

Provider Type ● Child Care Center ● Preschool Program ONLY



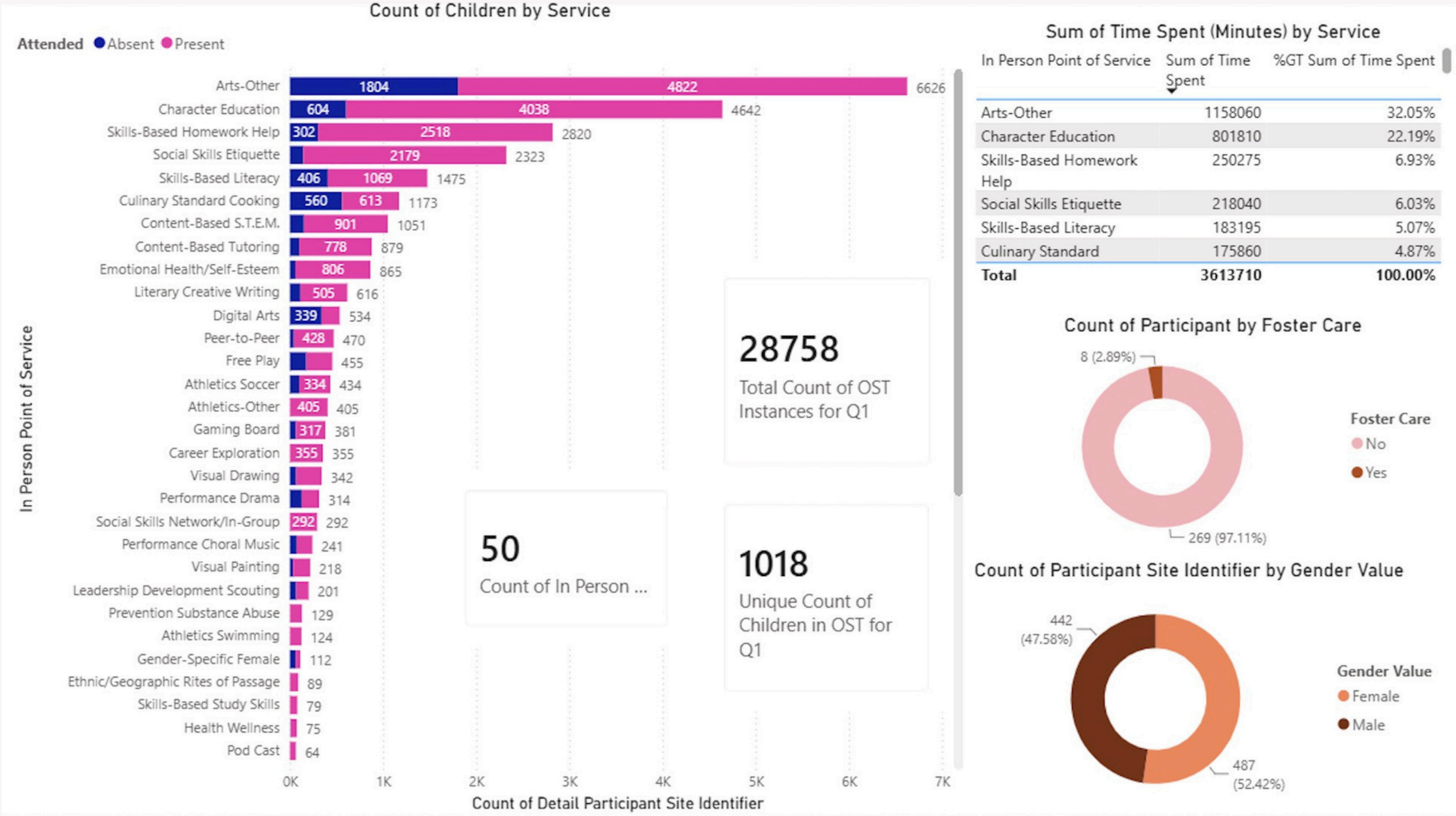
Average of Last TpfTScore by Site Name and Provider Type

Provider Type ● Child Care Center ● Preschool Program ONLY

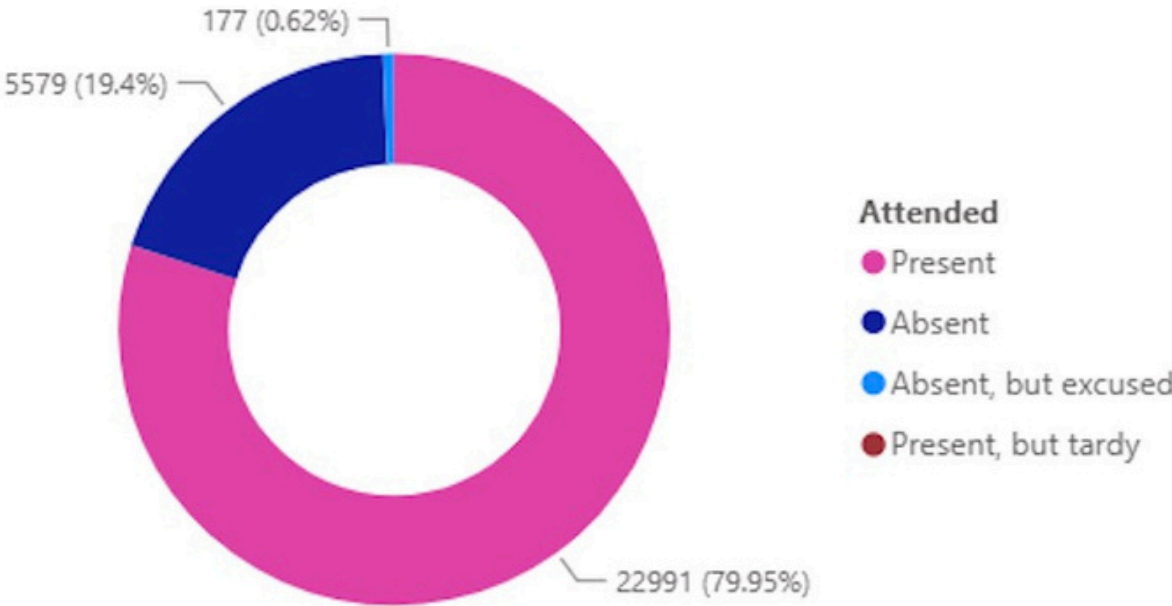


Part III: Out of School Time

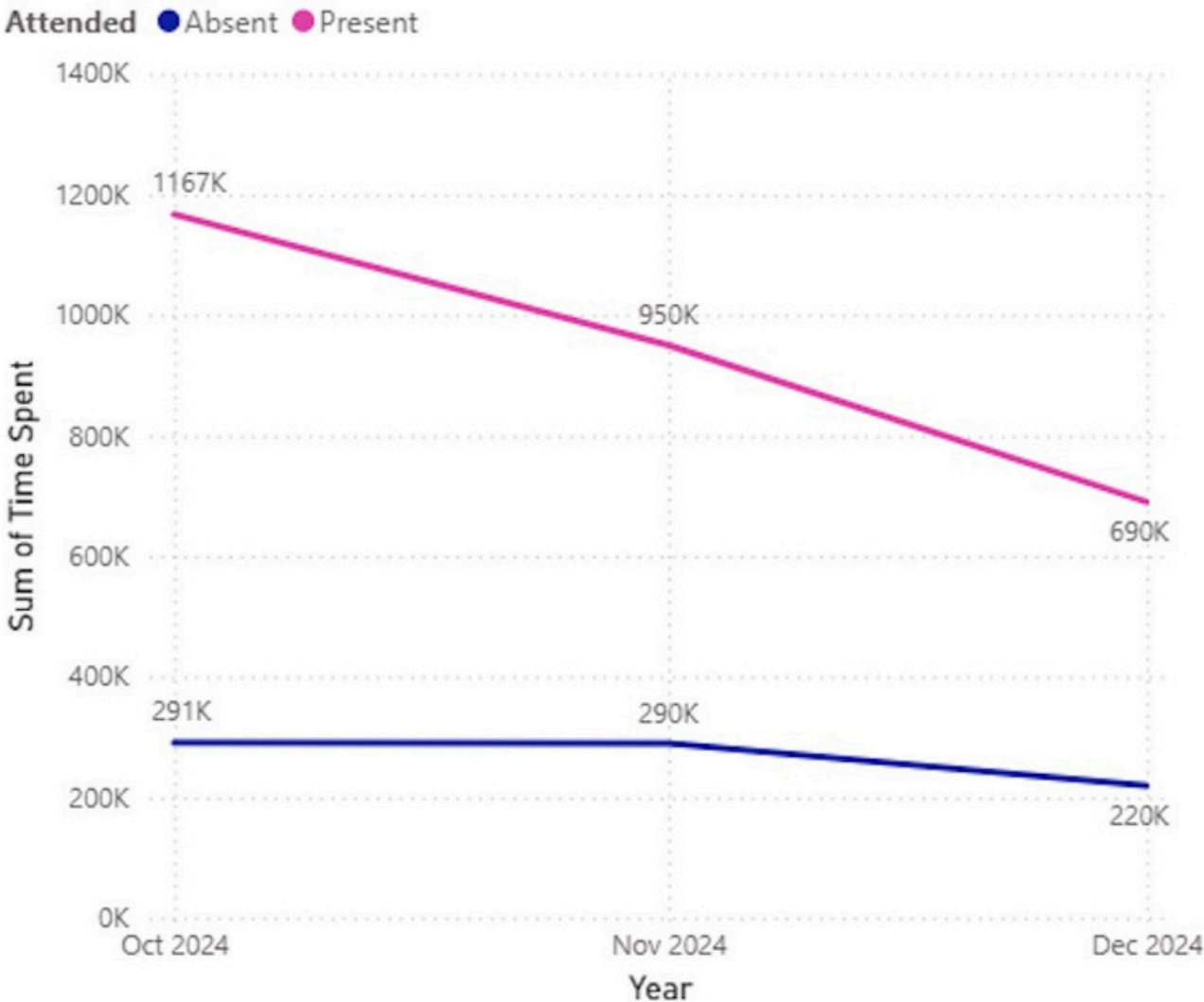
My goal with this part of the project was to determine where dependency on Out of School Time programs is the highest. I analyzed quarterly data first and then considered yearly outcomes.



Count of Participant by Attended



Sum of Time Spent in OST (Minutes) by Year, Month and Attendance



35

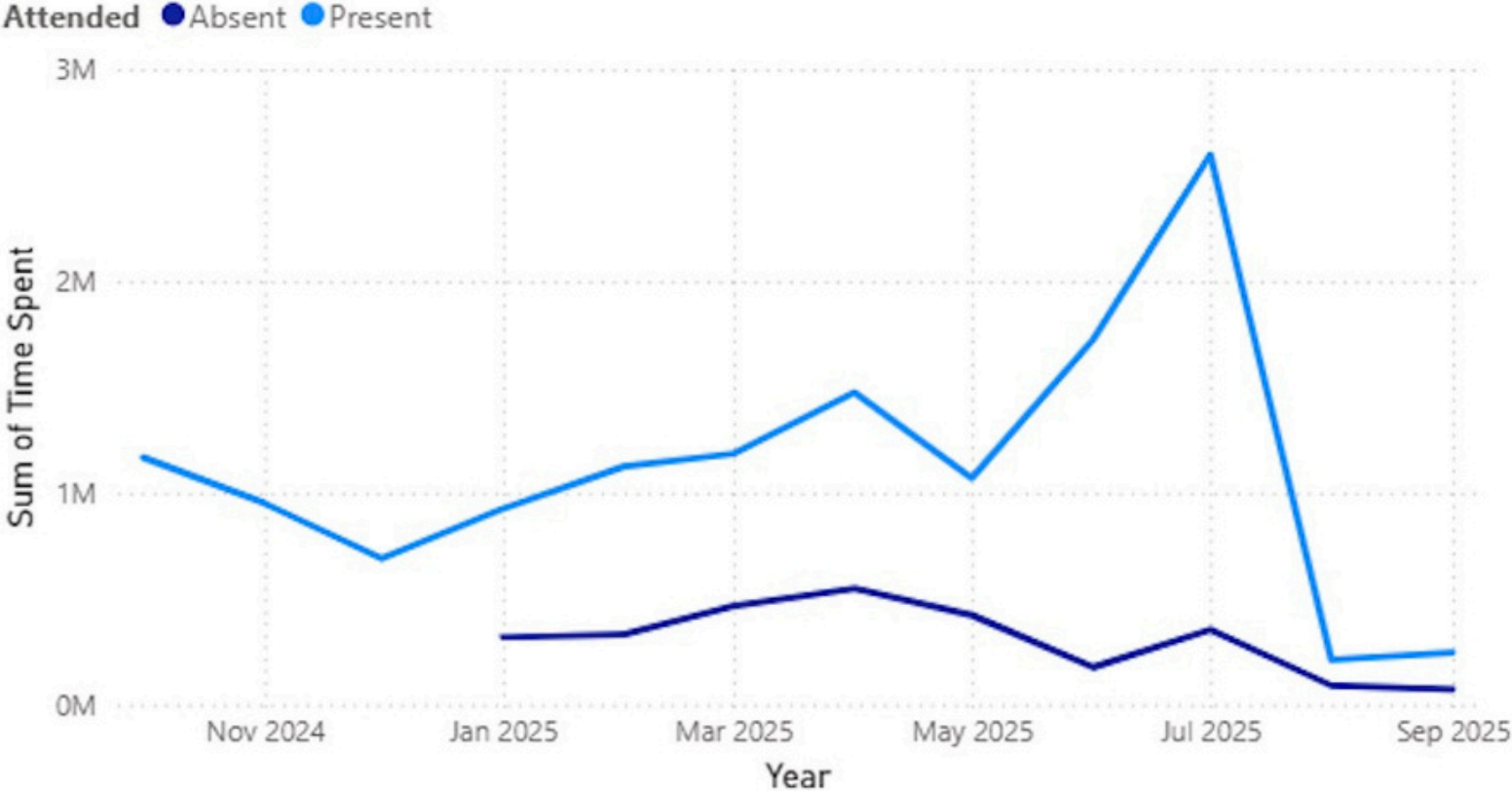
Count of Site Name

Children's OST Attendance Habits

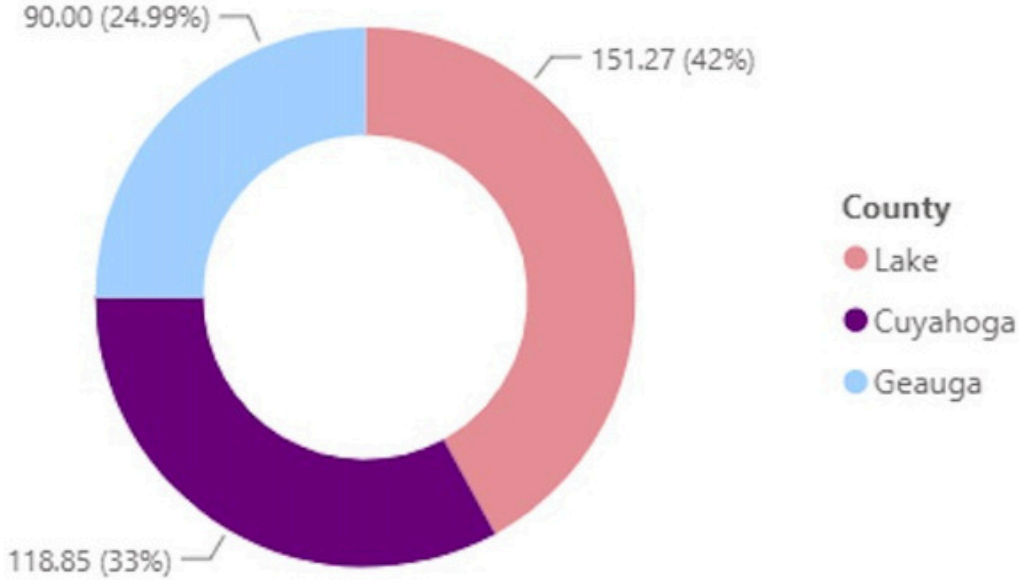
Attended	Count of Detail Participant Site Identifier	%GT Count of Detail Participant Site Identifier
Absent	5579	19.40%
Absent, but excused	177	0.62%
Present	22991	79.95%
Present, but tardy	11	0.04%
Total	28758	100.00%

The number of time spent at OST sessions started to decrease once the school year began for children who were present. The count started at about 1.2 million minutes and decreased to just under 700k by the end of the quarter for children who were actively attending OST. The minutes recorded for absent children decreased slightly by December.

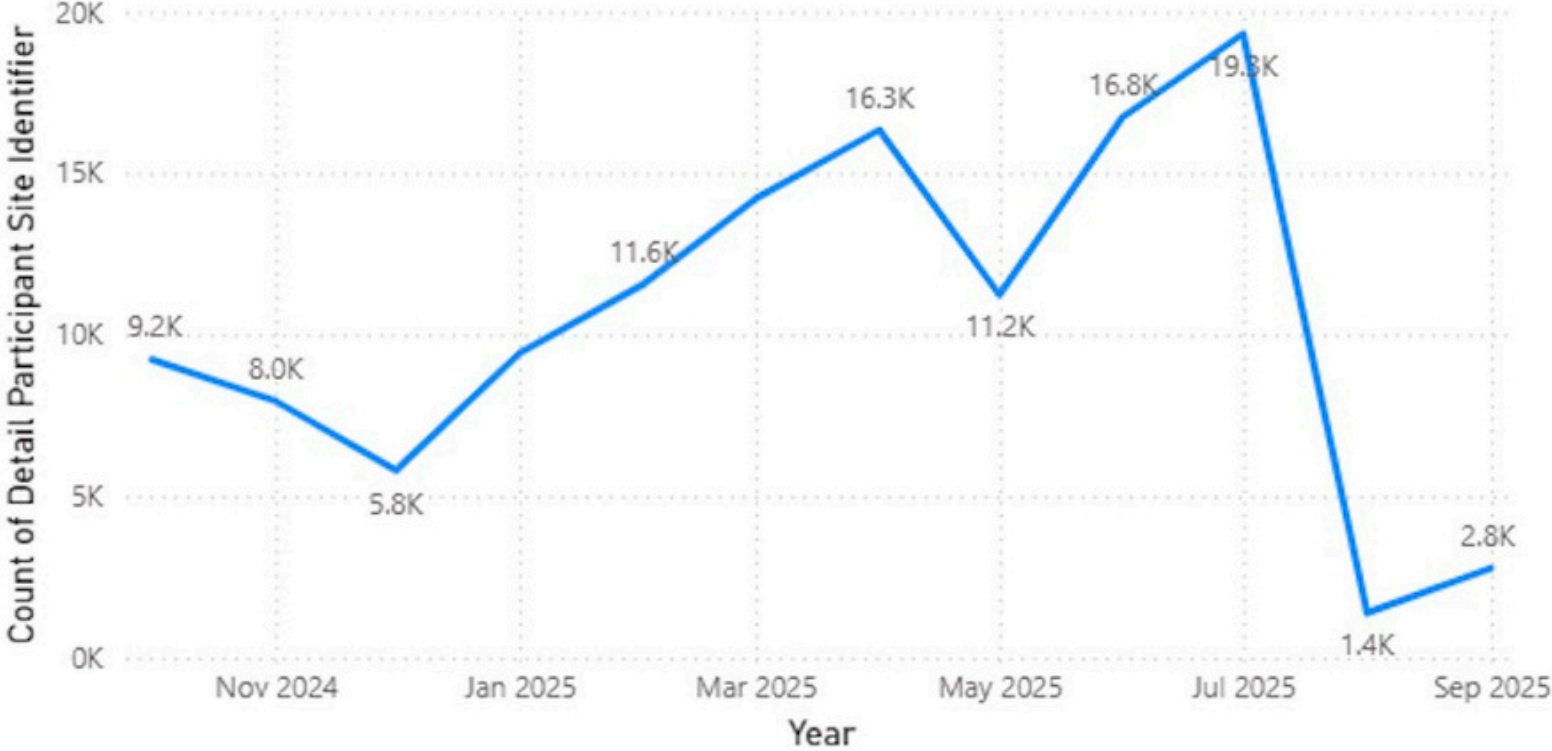
Count of Children in OST in 2024-2025 b Attendance



Average of Time Spent in OST by County



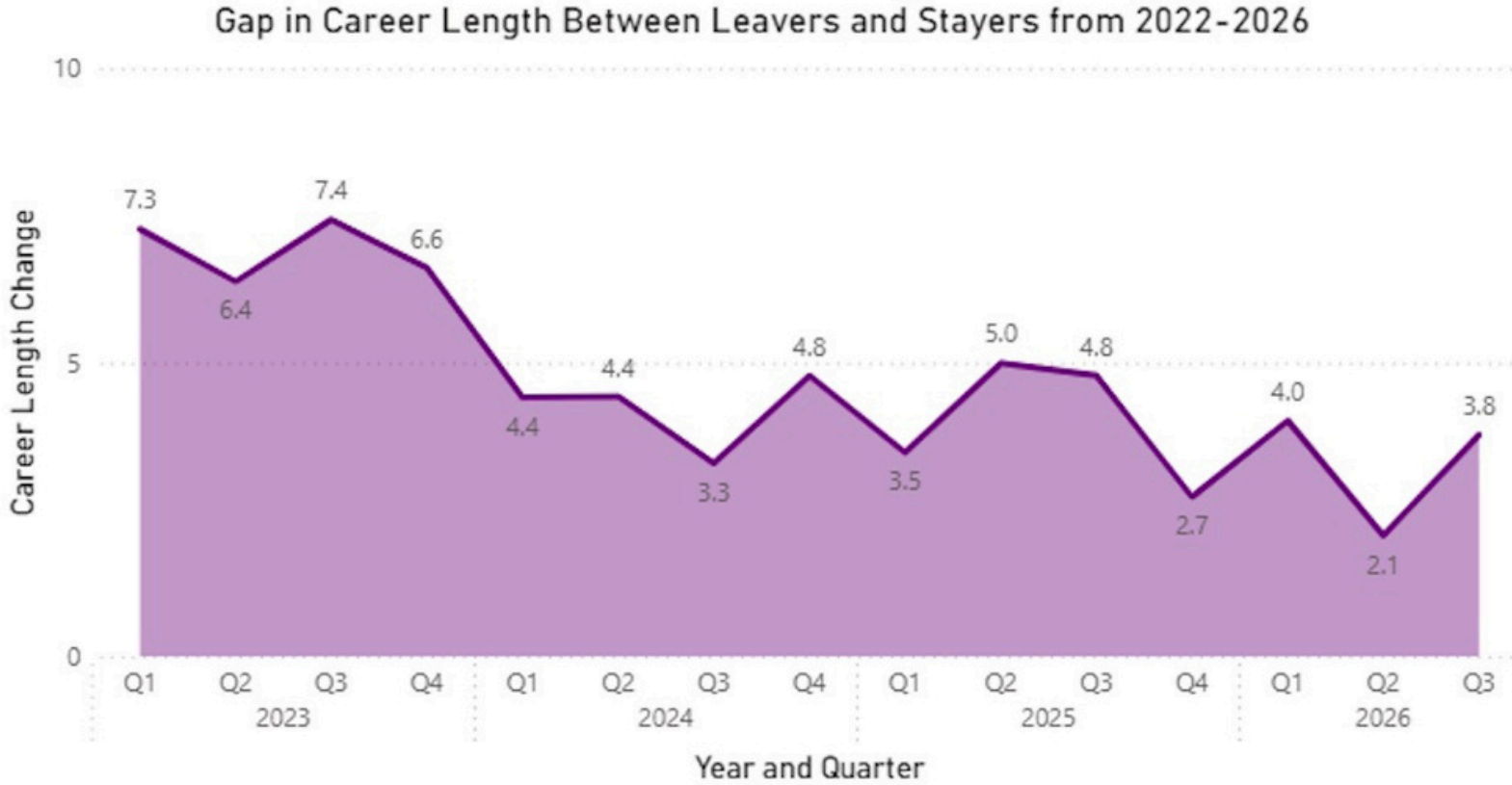
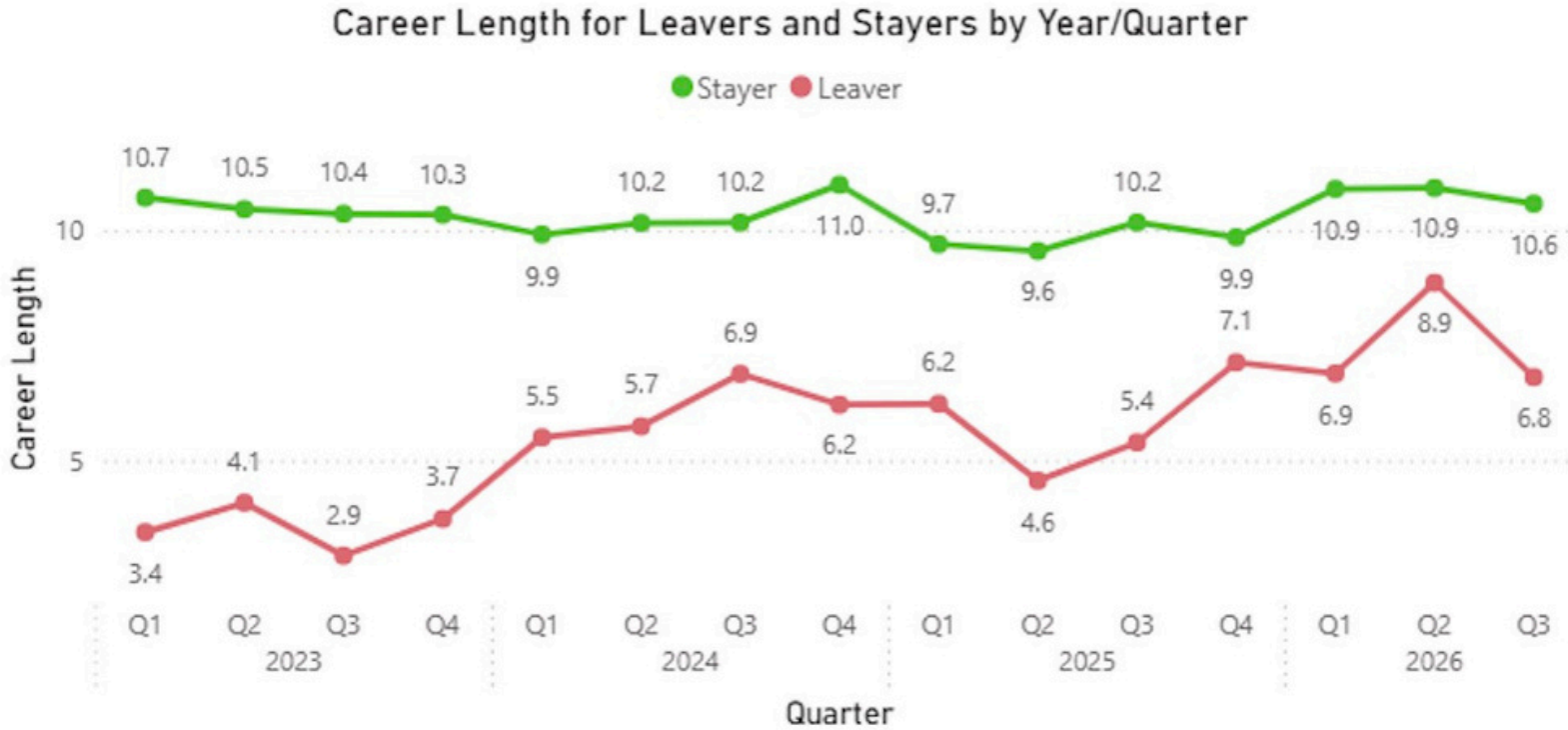
Count of Children in OST in 2024-2025



Primary Caregiver's Relationship to the Youth	Count of Detail Participant Site Identifier	%GT Count of Detail Participant Site Identifier
Mother	2000	81.60%
Father	273	11.14%
Grandmother	88	3.59%
Aunt	32	1.31%
Guardian	26	1.06%
Foster Parent	15	0.61%
Grandfather	10	0.41%
Stepmother	6	0.24%
Cousin	1	0.04%
Uncle	1	0.04%
Total	2451	100.00%

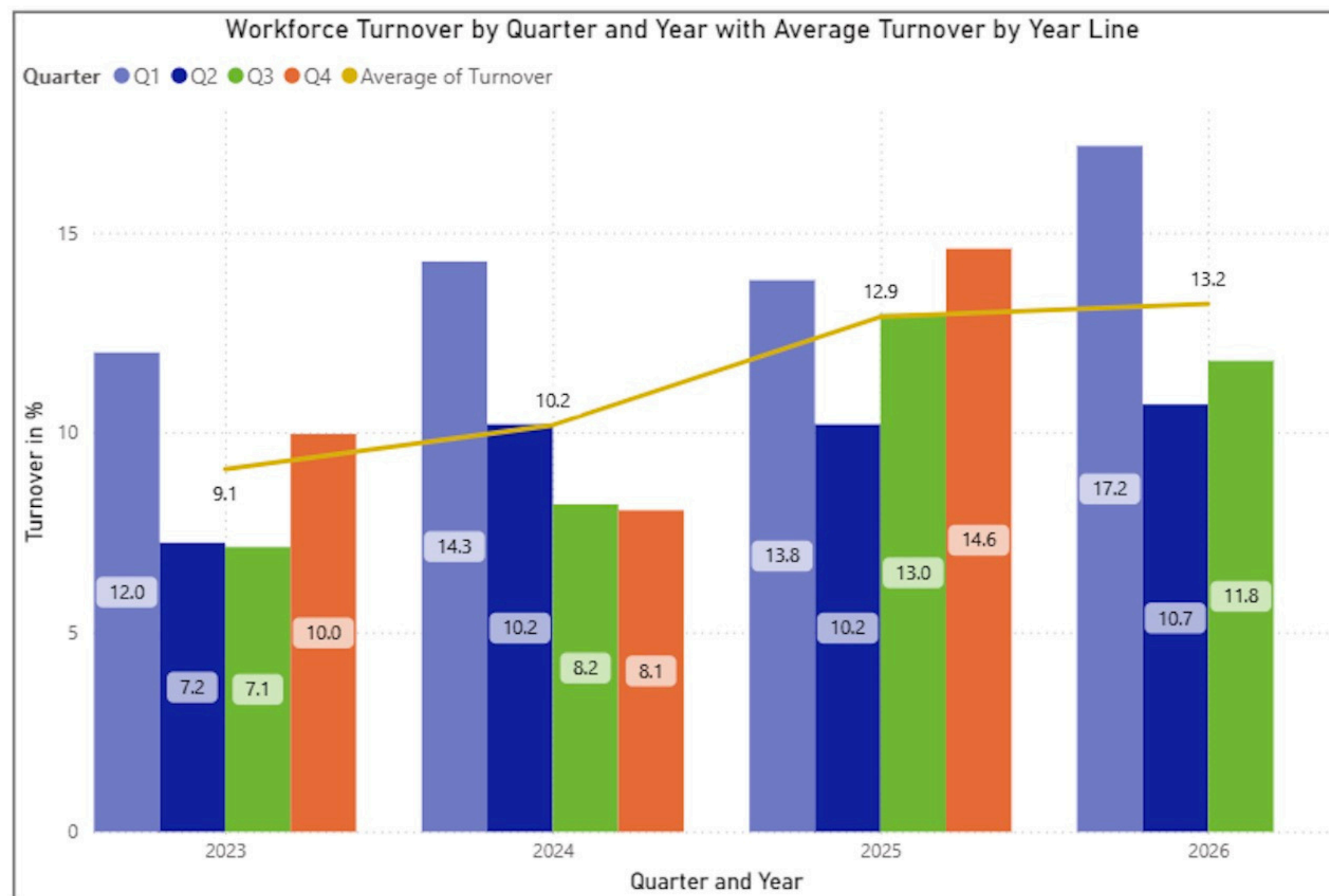
Part IV: Workforce Development

My goal with this part of the project was to better understand how the early childhood care workforce has changed over time.



Career Length of Leavers and Stayers with Turnover by Year/Quarter

Year	Quarter	Stayer (Career Length in Years)	Leaver (Career Length in Years)	Turnover %
2023	Q1	10.71	3.44	12.00
2023	Q2	10.46	4.08	7.24
2023	Q3	10.36	2.93	7.13
2023	Q4	10.34	3.73	9.96
2024	Q1	9.91	5.50	14.28
2024	Q2	10.16	5.74	10.20
2024	Q3	10.17	6.88	8.20
2024	Q4	10.99	6.21	8.05
2025	Q1	9.70	6.23	13.81
2025	Q2	9.55	4.56	10.20
2025	Q3	10.17	5.39	12.97
2025	Q4	9.85	7.13	14.60
2026	Q1	10.90	6.89	17.17
2026	Q2	10.92	8.86	10.70
2026	Q3	10.58	6.81	11.79





WORK TOGETHER!

Name of Project:
Data Analysis Portfolio

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